

S2 Text. Historical Case Analyses

0.1 Historical anchoring

The model predicts that codified systems with high legibility, high enforcement reward, institutional delegation, and restricted exit will produce enforcement-dominant dynamics through the endogenous feedback loop. This is a structural prediction about observable institutional configurations, and the historical record of codified systems provides the evidence base against which it is tested. The cases below are selected for documentation quality and structural diversity (religious and secular, Western and non-Western, medieval and modern). Each case is analyzed in terms of its code geometry and the observed enforcement dynamics.

The Inquisition (1231 to 1834). The Roman and Spanish Inquisitions represent the longest-running institutionalized enforcement apparatus in a codified moral system. The code geometry is explicit. Legibility was high: the Fourth Lateran Council (1215) mandated annual confession and communion, creating an auditable compliance metric. The Inquisitorial procedure (*inquisitio*) formalized deviation-detection through structured interrogation and denunciation networks [17]. Enforcement reward was institutionalized: inquisitors received a share of confiscated property; informers received immunity; cooperation yielded lighter sentences. Delegation was complete: enforcement authority resided in a specialized tribunal with trained personnel, procedural manuals (*Practica Inquisitionis* of Bernard Gui, c. 1323), and institutional continuity across generations. Exit was blocked by the conjunction of canon law (apostasy as capital offense), geographic immobility, and kinship embeddedness (entire families penalized for individual deviation). The model predicts capture for this configuration. The Inquisition produced a self-reinforcing enforcement apparatus that persisted for six centuries, concentrated punishment in a specialized minority, and systematically darkened the portrayal of alternatives (heresy as satanic contagion requiring quarantine). The temporal sequence matches the model's causal chain: institutional delegation preceded the intensification of outside-world degradation narratives. The early Inquisition (13th century) was a procedural innovation in enforcement; the elaborate demonological framework that cast heresy

as existential contamination developed later, as the enforcement institution consolidated communicative authority.

Post-Reformation confessionalization (16th to 17th centuries). After the Reformation, both Protestant and Catholic polities developed parallel disciplinary systems, a phenomenon historians term "confessionalization" [34,36]. The structural convergence is precisely what the model predicts. Lutheran, Calvinist, and Counter-Reformation Catholic authorities faced the same governance problem (maintaining doctrinal uniformity in a competitive religious market) and deployed the same structural instruments: legible compliance markers (mandatory church attendance records, catechism examinations, public confession schedules), delegated enforcement (church courts, consistories, visitation commissions), enforcement reward (ecclesiastical appointment and political preferment contingent on orthodoxy policing), and exit restriction (*cuius regio eius religio* eliminated denominational choice within territorial units). That three doctrinally incompatible systems independently converged on the same enforcement architecture is direct evidence that the dynamics are driven by code geometry rather than by theological content.

The Wahhabi-Saudi system (1744 to present). The pact between Muhammad ibn ʿAbd al-Wahhab and Muhammad ibn Saud coupled a doctrinal reform program with state-building capacity. Legibility was maximized by the Wahhabi insistence on observable orthopraxy: correct prayer form, avoidance of shrine visitation, public Quran recitation, visible rejection of practices classified as *bid'a* (innovation). Enforcement reward was fused with political loyalty: supporting the enforcement program was a condition of tribal alliance with the Saudi state. The Committee for the Promotion of Virtue and Prevention of Vice (CPVPV, formally established 1940, antecedents earlier) institutionalized delegation with street-level enforcement authority, arrest powers, and judicial backing. Exit was constrained by geographic isolation, tribal kinship structure, and the legal equation of apostasy with capital offense under Saudi jurisprudence. The model predicts enforcement concentration, endogenous narrative drift, and population retention. The historical record confirms all three. The CPVPV concentrated enforcement in a specialized cadre operating with state sanction. The Wahhabi doctrinal apparatus classified non-Wahhabi Muslim practice as *shirk* (poly-

theism), placing virtually all alternatives in the category of existential spiritual danger. Population retention was near-total in the core Najdi population throughout the period of CPVPV dominance.

The post-2016 Saudi reforms provide a natural test of the model's structural prediction. The curtailment of CPVPV enforcement powers reduced the delegation parameter. The simultaneous relaxation of dress enforcement and expansion of entertainment options reduced legibility and increased perceived outside-option quality. The model predicts regime transition toward mixed or quiet dynamics. Early indicators (increased public behavioral diversity, reduced religious policing, growth of a visible secular-leisure sector) are consistent with this prediction, though the transition remains incomplete.

Marxism-Leninism (1917 to 1991). The Soviet system provides the critical non-religious test case. If code geometry rather than theological content determines enforcement regime, a secular codified system with equivalent structural parameters should produce equivalent dynamics. Democratic centralism formalized legibility: party members demonstrated ideological conformity through public self-criticism sessions, denunciation of deviationists, and visible participation in party rituals [27]. Enforcement reward was extreme: career advancement, housing allocation, access to consumer goods, and physical safety were contingent on demonstrated compliance and active participation in enforcement [14]. Delegation was layered: the NKVD, party control commissions, and workplace party cells constituted a specialized enforcement apparatus with trained personnel, procedural protocols, and institutional continuity. Exit was blocked by internal passport systems, criminalized emigration, and total dependence of economic life on state employment.

The model predicts capture. The Soviet system produced a self-reinforcing enforcement apparatus in which punishment was concentrated in a specialized minority (party organs and security services), the outside world was systematically portrayed as hostile and decomposing (capitalist encirclement, bourgeois contamination; Yurchak 2005), and population retention was near-total. The Great Purge (1936 to 1938) exemplifies the model's $\sigma \times \pi$ interaction at low legibility: genuine political loyalty was opaque (low σ), but the reward for denunciation (career advancement, immunity, apartment allocation) was so high that enforcement was vigorous despite massive false-positive

rates. The model reproduces this dynamic at $\sigma = 0.25$, $\pi = 0.50$ (§8.9).

The Soviet collapse (1989 to 1991) maps to the model's prediction about exit-cost reduction. When the enforcement apparatus lost coercive capacity (Gorbachev's refusal to deploy force, withdrawal of military backing from satellite states), the exit parameter shifted. The model predicts that capture regimes with suddenly reduced exit costs transition to collapse. The speed of Soviet-bloc dissolution (months, not decades) is consistent: the population was retained by structural coercion, not by attachment, and when coercion failed, exit was immediate.

Quiet-regime confirmations. The model predicts that systems with low legibility and low substitutability will not produce enforcement-dominant dynamics regardless of doctrinal stringency. The Theravāda forest tradition emphasizes internal meditative attainment (*jhāna*, insight stages) that is opaque to external observation. A meditator's progress is not legible to others; no external performance substitutes for internal realization. The tradition has not produced an enforcement apparatus despite demanding behavioral requirements (the 227 *pātimokkha* rules of the monastic code). The *pātimokkha* rules are legible (observable behavioral compliance), but the tradition's core soteriological metric (meditative attainment) is not. Enforcement operates on the behavioral rules (monastic discipline committees exist), but it cannot extend to the tradition's central evaluative criterion. The enforcement apparatus remains limited in scope because the code's core dimension resists legibility.

Contemporary Quaker meetings present an even cleaner case. The tradition's central criterion (responsiveness to the Inner Light) is entirely opaque to external observation. No behavioral proxy substitutes for the internal state. The model predicts quiet dynamics. Contemporary Quaker meetings confirm the prediction. The contrast with historical Quakerism is informative: when 18th-century Quaker practice included legible compliance markers (plain dress, plain speech, endogamy requirements), enforcement was active and disownment was common. As these markers were relaxed across the 19th and 20th centuries, enforcement ceased. The trajectory tracks the legibility parameter, not doctrinal intensity.

The boundary case. The contemporary Haredi community in Bnei Brak scores high on leg-

ibility, substitutability, and enforcement affordance but moderate on centralization (enforcement authority is distributed across multiple competing rabbinical courts and communal leaders rather than monopolized by a single body). The model predicts mixed-to-capture boundary dynamics. The observed reality matches: Haredi communities exhibit vigorous enforcement (social sanctions for dress-code violations, intermarriage, Shabbat desecration) and substantial outside-option degradation (secular education discouraged, economic life substantially internal, family severance for defectors), but the fragmentation of enforcement authority across competing jurisdictions limits the consolidation that full capture requires.

The cross-traditional pattern. Across the Inquisition, the Wahhabi-Saudi system, the Soviet party apparatus, post-Reformation confessionalization, and the CCP discipline system, the same five-element conjunction appears: legible compliance metrics, positive enforcement reward, institutional delegation to a specialized minority, compounding enforcement capacity (records, networks, procedural precedent), and restricted exit. Where this conjunction is present, enforcement-dominant dynamics emerged regardless of whether the codified system was Christian, Islamic, or Marxist, and regardless of the specific doctrinal content enforced. Where the conjunction is absent (forest monasticism, Quaker meetings), enforcement-dominant dynamics did not emerge despite demanding normative expectations. The recurrence is not coincidence. It is structural convergence produced by equivalent code geometries operating on equivalent institutional substrates.